ModelIt!

Teacher's Guide

How Do Maps Help Us Find Things?

Mapping the Land, Water, and Features of Earth

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-ESS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-ESS2-2

Develop a model to represent the shapes and kinds of land and bodies of water in an area.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model (map) to represent the shapes and kinds of land and bodies of water in an area.
Disciplinary Core Idea	**ESS2.B Plate Tectonics and Large-Scale System Interactions** — Maps show where things are located. One can map the shapes and kinds of land and water in any area.
Crosscutting Concept	**Patterns** — Students identify patterns in where water collects (low areas) and where land rises (high areas) on maps.

CAST Testing Alignment

MULTIPLE CHOICE:

On a map, you see a blue winding line between two brown mountain shapes. What does the blue line most likely represent?

CONSTRUCTED RESPONSE:

Explain how a map helps us understand a place we have never visited. What can a map show us about the land and water?

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Maps are one of the oldest and most useful tools humans have ever created. From ancient cave drawings to satellite images, maps help us understand the shape of our world and find our way around it.

What Maps Show Us

Maps are flat representations of the Earth's surface, showing features from a bird's eye view. They use colors, lines, and symbols to represent real things. Blue usually shows water — oceans, rivers, and lakes. Green often shows forests or low-lying land. Brown shows mountains or deserts. Maps can show a small area like your neighborhood or a whole continent.

Landforms and Bodies of Water

Earth's surface has many different shapes. Mountains are tall peaks of land that rise high above the surrounding area. Valleys are low areas between mountains. Plains are large flat areas of land. Rivers are flowing bodies of water that travel from high areas to low areas. Lakes are bodies of water surrounded by land. Oceans are the largest bodies of water on Earth.

How Water Shapes the Land

Water always flows downhill because of gravity. Rain falls on mountains and flows down into valleys, creating streams and rivers. Rivers carry water to the lowest points, where it collects in lakes or flows all the way to the ocean. Over millions of years, water carves valleys, creates canyons, and shapes the land we see today.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify land height as an external component and water location and map features as internal components that change based on the shape of the terrain.

E **Establish Relationships**
Students determine that land height affects where water flows — water moves from high areas to low areas (negative relationship between land height and water accumulation at that spot).

V **Visualize & Model**
Students build a model (map) showing how different land heights create different features and determine where water collects.

E **Evaluate Outcomes**
Students run mountain, flat plains, and valley scenarios to see how different land shapes create different maps and water patterns.

R **Revise & Extend**
Students add map scale or map symbols to explore how maps communicate additional information about the land.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever looked at a map? What did you see on it? Today we are going to learn how maps show us what the Earth looks like!"*

Watch For

Show a simple map and a satellite photo of the same area. Ask: Can you see the same things in both?

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to learn to read maps AND build our own 3D map!"*

Watch For

Read objectives together. Introduce map, landform, and body of water with pictures.

Slide 3: The Big Question

3 min

Say: *"How can a piece of paper or a screen show you what a mountain or a river looks like? Maps are like magic pictures of the Earth!"*

Watch For

Show a photo of a mountain and then the same mountain on a map. Ask: How did the mapmaker show the mountain?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to understand how the shape of the land decides where water goes."*

Watch For

Preview LEVER steps. Show crumpled paper as a quick 3D land model — pour water on it to show flow.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What are the parts of our mapping system? Land Height is something nature built. Water and Map Features change based on the land."*

Watch For

Guide sorting: Land Height is external, Water Location and Map Features are internal.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When the land is very tall like a mountain, where does the water go? Up or down?"*

Watch For

Draw arrows showing Land Height affects Water Location (water flows to low areas). Discuss how maps show these features.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test three areas: a mountain, flat plains, and a valley. Where will the water be in each?"*

Watch For

Students predict then run simulations. Compare how each area looks on a map. Use clay models if possible.

Slide 8: What Did We Discover?

7 min

Say: *"Now you can read a map! Blue means water, and water always flows to the lowest spot. Maps show us the world from above!"*

Watch For

Give each student a simple map to read. Ask them to find the mountains, rivers, and flat areas.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Land Height
- **INTERNAL:** Water Location, Map Features

Relationships

- **Land Height to Water Location:** NEGATIVE (-) — When land is high (mountains), water does not stay there — it flows downhill. Water collects in the lowest areas like valleys, lakes, and rivers. So tall land means less water at the top and more water at the bottom.
- **Land Height to Map Features:** POSITIVE (+) — Different land heights create different map features. Mountains show as brown bumpy shapes, valleys show as low areas between peaks, and flat plains show as smooth green areas. The more varied the land height, the more interesting the map.

Key Discoveries

- Maps use colors and symbols to represent real land and water features from above
- Water flows downhill, so it collects in the lowest parts of the land — valleys, lakes, and rivers
- Mountains appear on maps with special markings because they are the highest points
- Maps help us understand places we cannot see or visit by showing the shapes of land and water

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast

Full period available	Component Jigsaw — deep collaboration, every student contributes
After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Maps show exactly what you see from the ground"	Maps show a bird's eye view — from above. They look very different from what you see standing on the ground. You cannot see a river winding for miles when you stand next to it, but on a map you can see its entire path from start to end.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Blue on a map means the sky"	Blue on most maps represents water — oceans, rivers, lakes, and streams. Maps show the Earth's surface, not the sky. When you see blue on a map, think water!
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Mountains are always covered in snow"	Not all mountains have snow! Only the very tallest mountains are cold enough at the top for snow to stay year-round. Many mountains are covered in forests, grass, or bare rock. The height and location of the mountain determines what is on top.